

Endpoint on Higgs Structural Identity: Bridge-Carrier Occupation and No Fifth Identity Ground

Paper 2 of the Higgs Structural Endpoint Arc

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Abstract

This manuscript proves the second paper in the Higgs structural endpoint arc. Paper 1 established the Higgs bridge/fixation carrier

$$C_H := H_{\text{br}/\text{fix}}$$

as the standing-bearing carrier forced by a neutral Higgs-domain order-to-deformation and mass-response target. The present paper asks the next question: what fixes Higgs identity once the carrier has been admitted? The answer is an endpoint theorem, not a definition by stipulation. In a same-scope Higgs-domain regime preserving the inherited target D_H , the inherited role bundle H_{role} , admissible redescription, and the Paper-1 normal form R_H , every admissible Higgs identity claim must factor through C_H . Scalarity, vev-status, potential-minimum status, pole-mass status, RG-running status, Yukawa-form status, mass-matrix status, vacuum-branch status, thermal-transition status, and metric-comparison status are identity representatives or downstream route statuses; none supplies primitive Higgs identity by itself.

The proof proceeds by separating identity grounds from descriptions, then exhausting the same-scope identity alternatives. A Higgs identity claim is a claim that purports to determine what counts as the Higgs in D_H , rather than merely describing, measuring, coordinating, or routing Higgs-associated structure. The representative catalogue is admitted as useful physics language, but each entry fails as primitive identity ground unless it presupposes or factors through the bridge carrier. The identity normal form

$$R_H^{\text{id}} = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H^{\text{id}}, \rho_H^{\text{id}})$$

records that admissible identity-bearing use must preserve the same target, the same carrier occupation, the same role bundle, and the same admissibility boundary. The ordered fail-closed identity classifier places every identity candidate into one of six classes: `CarrierIdentical`, `RepresentativeOnly`, `RouteStatus`, `StandingSmuggling`, `RoleIncomplete`, or `TargetUpdate`. Carrier-identical candidates are the same identity role under licensed redescription; representative and route candidates do not ground identity; standing-smuggling is blocked by the no-generator/no-transfer/no-repair discipline; role-incomplete candidates fail the target; target updates leave the same-scope identity theorem.

The main theorem is the *Higgs Structural Identity Theorem*:

$$\text{Id}_H = \text{unique admissible occupation of } C_H \text{ inside } D_H.$$

Equivalently, any same-scope occupant satisfying the full Higgs identity role is not a second primitive identity ground but the same bridge/fixation carrier up to role-preserving redescription. No fifth identity ground remains outside carrier occupation, representatives, downstream routes, standing smuggling, role incompleteness, and target update. The result preserves ordinary Standard Model Higgs language as representative language while denying its promotion into primitive identity authority.

Keywords: AASC; Higgs structural identity; Higgs bridge; carrier occupation; representative collapse; no fifth identity ground; Standard Model; identity-preserving redescription; endpoint theorem; same-scope exhaustion.

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Mathematical-status lock and proof-class boundary

This paper is an AASC fixed-target identity theorem. It is not a new Higgs-sector model, not a claim that the Higgs is not a scalar in Standard Model presentations, not a collider phenomenology paper, not a derivation of the Higgs vev or pole mass, not a hierarchy solution, not a vacuum-stability calculation, and not a classification of all possible BSM sectors. Standard electroweak language supplies the comparison arena. AASC supplies the status calculus that separates identity grounds, representatives, measurements, downstream routes, and richer-scope updates.

The proof is evaluated downstream of Paper 1. The inherited Paper-1 package is

$$\mathcal{P}_H^{(1)} = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, R_H, \mathbf{SF}_H, E_H^{(1)}), \quad C_H = H_{\text{br/fix}}.$$

Paper 1 imported the four-role kernel $K_D = (K, \simeq_{\text{gov}}, I_D)$, $K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$, and specialized its consequences to electroweak mass-response. The present paper does not rederive the kernel. It asks what identity theorem follows once C_H is available as the carrier forced by the Higgs-domain target.

A same-target objection must invalidate one of the following: the inherited carrier package, the definition of a Higgs identity claim, the identity/description separation, the representative failure certificates, the identity normal form, the same-scope replacement theorem, the no-fifth-identity-ground theorem, or the final structural identity endpoint. A preference for scalar-first language or a model-building description does not by itself contest the theorem unless it supplies a same-scope primitive identity ground that preserves D_H , H_{role} , and the admissibility boundary without reducing to C_H .

Read this first: identity proof spine

The proof has five layers.

Layer I: inherited carrier package. Paper 1 establishes that a non-degenerate Higgs-domain order-to-deformation and mass-response regime forces a bridge-carrier normal form and imports bridge-witness closure identifying the occupant as $C_H = H_{\text{br/fix}}$. This paper treats that package as the local input, not as material to be reproved.

Layer II: identity claim typing. A Higgs identity claim is stronger than a description, measurement, route, coordinate, or representative. It purports to determine what counts as the Higgs in the same target D_H . Therefore the claim must preserve target identity, the carrier role, the role bundle H_{role} , and standing-preserving redescription.

Layer III: representative identity collapse. Scalarity, vev-status, potential-minimum status, pole-mass status, running-parameter status, Yukawa-form status, mass-matrix status, vacuum-branch status, thermal-transition status, and metric-comparison status are admitted as representative or downstream descriptions. They do not independently preserve the full identity role. Their lawful identity-bearing use is mediated by C_H .

Layer IV: classifier totality, identity normal form, and replacement exhaustion. Every same-scope Higgs identity candidate is classified by ρ_H^{id} into `CarrierIdentical`, `RepresentativeOnly`, `RouteStatus`, `StandingSmuggling`, `RoleIncomplete`, or `TargetUpdate`. Only admissible standing-bearing identity grounds factor through the identity normal form; rejected candidates receive a closed failure or downstream status. The only standing-bearing same-scope identity class is `CarrierIdentical`, and that class is role-equivalent to C_H .

Layer V: structural identity endpoint. The endpoint is

$$\text{ld}_H = \text{Occ}_{D_H}(C_H),$$

unique admissible occupation of the Higgs bridge/fixation carrier role. No fifth same-scope identity ground remains.

Dependency-inversion exclusion

The dependency order is not

$$\text{Higgs descriptor} \Rightarrow \text{Higgs identity}.$$

The dependency order is

$$\text{Higgs-domain target} \Rightarrow \text{carrier necessity} \Rightarrow \text{identity endpoint}.$$

A representative cannot define the identity rule by which it becomes the Higgs. Once the same-scope target and carrier package are fixed, a submitted identity ground must either preserve C_H , reduce to a licensed representative of C_H , function as a downstream route, declare target update, or fail.

1 Introduction: from carrier to identity

Paper 1 closed the local carrier question. It did not merely name a Higgs field or import a Standard Model scalar presentation. It began from a neutral Higgs-domain target in which gauge/representation order had to be used for deformation and mass-response claims without target drift, hidden selector import, or representative promotion. From that target it proved bridge-carrier necessity and identified the imported bridge-witness occupant as $C_H = H_{\text{br/fix}}$. That result supplies the local standing-bearing carrier for the arc.

A carrier theorem is distinct from an identity theorem. A carrier theorem says which role must be occupied for Higgs-domain standing to be possible. An identity theorem says what follows for claims that purport to determine what the Higgs is. This manuscript proves that second theorem.

The ordinary Higgs lexicon offers many possible identity temptations:

$$\text{scalar field, } v, \quad V(H), \quad m_H, \quad \lambda(\mu), \quad y_f, \quad \text{vacuum branch, } T\text{-profile, } \text{metric scale}.$$

Each may be legitimate in the appropriate context. None, merely by being written, measured, or used in a calculation, fixes Higgs identity. The reason is structural: identity-bearing use must preserve the same target, the same admissibility boundary, the same role bundle, and the same carrier occupation. A representative can display a carrier only after the carrier role has standing; it cannot supply that standing by itself.

The central theorem therefore has a deliberately spare form:

$$\text{ld}_H = \text{unique admissible occupation of } C_H.$$

This does not deny that the Standard Model Higgs is represented by a scalar doublet, that electroweak input values are physically measured, that Higgs excitation has a pole mass, or that vacuum and thermal calculations are meaningful. It denies a different claim: that any of those representative features is the primitive identity ground of the Higgs in the fixed target.

1.1 What the paper proves

The paper proves five outputs:

1. identity-bearing Higgs claims are stricter than Higgs descriptions, measurements, or routes;
2. every familiar representative candidate fails as primitive identity ground;
3. every submitted Higgs identity candidate receives a fail-closed classifier status, and every standing-bearing same-scope Higgs identity ground factors through R_H^{id} ;
4. every same-scope replacement satisfying the full identity role reduces to the same carrier occupation under role-preserving redescription;
5. no fifth identity ground remains.

The identity endpoint is structural. It is not a numerical prediction and not a replacement for phenomenological Higgs-sector descriptions.

1.2 What the paper does not prove

Nonclaim 1.1 (No numerical Higgs prediction). The paper does not derive the Higgs vev, Higgs pole mass, self-coupling, total width, rare decay rate, Yukawa matrix, electroweak input value, hierarchy magnitude, or vacuum lifetime.

Nonclaim 1.2 (No denial of scalar representation). The paper does not deny the scalar-field representation of the Higgs in Standard Model QFT. It denies only that scalarity is the primitive identity ground once the target is a same-scope identity claim.

Nonclaim 1.3 (No global BSM exclusion). The paper does not rule out additional scalars, composite Higgs models, supersymmetry, portal sectors, or other extensions. It classifies their relation to same-scope Higgs identity: carrier-preserving enrichment, representative duplication, role split, target update, or failure.

2 Inherited package from Paper 1

Paper 1 supplies the local package used below. The present manuscript restates the package so that the identity theorem is independently navigable, while leaving the kernel import and carrier proof in their proper source.

Definition 2.1 (Inherited Higgs carrier package). The inherited Paper-1 package is

$$\mathcal{P}_H^{(1)} = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, R_H, \text{SF}_H, E_H^{(1)}),$$

where:

- D_H is the neutral Higgs-domain order-to-deformation and mass-response target;
- $C_H = H_{\text{br/fix}}$ is the Higgs bridge/fixation carrier;
- H_{role} is the role bundle

$$(\text{Gaugelnh}, \mathcal{N}^{\text{res}}, \text{VacSupp}, \text{DefMed}, \text{MassResp}, \text{IdCont}, \text{StandTrans}, \text{RedStab});$$

- $\mathcal{R}_H$ is the catalogue of Higgs representatives;
- R_H is the local Higgs normal form for same-scope Higgs standing claims;
- SF_H is the standing filter separating primitive, representative, route, standing-smuggling, role-incomplete, and target-update claims;
- $E_H^{(1)}$ is the initial representative-exhaustion register.

Status 2.2 (Dependency posture). The kernel, admissibility structure, fixed-domain exhaustion, AMetric boundary, impossibility-suite consequences, bridge-witness closure, and Standard Model structural instantiation are imported through Paper 1. The present paper uses them only through $\mathcal{P}_H^{(1)}$.

Table 1: Paper-1 package items used in Paper 2

Item	Imported content	Use in identity theorem
D_H	Neutral Higgs-domain transition target	Fixes same-scope identity domain.
C_H	Closed Higgs bridge/fixation carrier	Supplies candidate identity occupant.
H_{role}	Gauge inheritance, residual/vacuum support, deformation mediation, mass-response readiness, identity continuity, standing transfer, redescription stability	Defines what an identity ground must preserve.
$\mathcal{R}_H$	Scalar, vev, potential, pole mass, RG, Yukawa, mass matrix, vacuum, thermal, metric descriptions	Provides representative catalogue to audit.
R_H	Local Higgs standing normal form	Basis for identity-specific normal form R_H^{id} .
SF_H	Same-scope standing classifier	Specializes to identity classifier ρ_H^{id} .

Lemma 2.3 (Carrier package sufficiency for identity analysis). *The inherited package $\mathcal{P}_H^{(1)}$ is sufficient to evaluate same-scope Higgs identity claims. No additional kernel import is required in the present paper.*

Proof. A same-scope identity claim must specify a target, the role bundle whose preservation fixes same-object status, the admitted representatives that may describe the object, and the standing filter that separates primitive, representative, route, failure, and target-update uses. The inherited package contains exactly these fields. The kernel import and admissibility consequences are needed to justify those fields; Paper 1 supplied that justification. Paper 2 can therefore work locally with the package rather than re-opening the kernel proof. $\square$

3 Identity claims versus descriptions

The most common failure in Higgs identity language is a type error: a successful description is treated as an identity ground. The manuscript therefore begins with strict typing.

Definition 3.1 (Higgs identity claim). For a fixed Higgs-domain target D_H , a Higgs identity claim is a claim

$$\text{IdClaim}_{D_H}(X)$$

asserting that X determines what counts as the Higgs in D_H , rather than merely describing, measuring, coordinating, routing, or extending Higgs-associated structure.

Definition 3.2 (Identity ground). An identity ground for the Higgs in D_H is a standing-bearing primitive G such that

$$\text{IdGround}_{D_H}(G) = 1$$

and same-scope Higgs identity claims are licensed only through preservation of G and the role bundle H_{role} . An identity ground must be reference-stable, standing-bearing, admissibility-preserving, and irreversibility-compatible.

Definition 3.3 (Representative description). A representative description is a scalar, coordinate, potential, field chart, measured response, coupling form, matrix entry, RG chart, vacuum presentation, thermal profile, metric projection, or other presentation that can lawfully describe C_H or a downstream route from C_H , but does not by itself determine the same-scope Higgs identity ground.

Definition 3.4 (Identity-role occupation). A candidate X supplies identity-role occupation in D_H , written

$$\text{IdRoleOcc}_{D_H}(X) = 1,$$

when X is submitted as the standing-bearing locus through which same-scope Higgs identity claims are licensed while preserving D_H , H_{role} , identity-preserving redescription, and the admissibility boundary. This definition does not identify X with C_H ; it states the neutral role that any submitted identity ground must occupy.

Criterion 3.5 (Identity-ground criterion). A candidate X has primitive Higgs identity standing only if all five conditions hold:

1. X preserves D_H rather than changing target;
2. X preserves the full role bundle H_{role} ;
3. X is invariant under same-scope identity-preserving redescription;
4. X does not derive its authority from an empirical representative, route success, downstream fit, or hidden selector;
5. X supplies identity-role occupation rather than representative availability.

Lemma 3.6 (Identity-role occupation is neutral at admission). *The identity-ground criterion does not select C_H by definition. It admits any candidate that supplies the full same-scope identity role, then leaves the identity-normal-form and replacement arguments to decide whether that role has more than one occupant.*

Proof. The criterion mentions target preservation, role preservation, redescription invariance, no hidden authority, and identity-role occupation. None of these conditions names C_H as the selected identity ground. A candidate satisfying them still must pass through the classifier, the replacement theorem, and the no-fifth-ground theorem. The conclusion $\text{Id}_H = \text{Occ}_{D_H}(C_H)$ is therefore not present at the criterion stage. $\square$

Lemma 3.7 (Identity-description separation). *Descriptive adequacy does not imply identity-grounding standing:*

$$\text{Describes}_{D_H}(X, C_H) = 1 \not\Rightarrow \text{IdGround}_{D_H}(X) = 1.$$

Proof. A description may be faithful to a carrier while lacking carrier priority. For example, a coordinate can describe a standing object, a pole mass can measure an excitation, and a potential can present a model chart. None of these facts ensures that the description itself preserves target identity, supplies the full role bundle, or blocks same-act repair and hidden selector import. Identity-grounding standing is stricter than descriptive success. Therefore descriptive adequacy does not entail identity standing. $\square$

Corollary 3.8 (No identity from successful measurement). *A measured Higgs-associated quantity may be a valid empirical witness without becoming a primitive Higgs identity ground.*

Proof. A measurement can fix a representative or response value. Identity grounding requires the standing-bearing role by which such measurements are measurements of the Higgs-domain carrier. The measurement presupposes the identity target; it does not create it. $\square$

Remark 3.9 (Criterion codes for representative audit). The representative tables use the following criterion codes: C1 target preservation, C2 full role-bundle preservation, C3 identity-redescription invariance, C4 no hidden empirical/route/selector authority, and C5 identity-role occupation.

4 Representative identity catalogue

The representatives below are admitted into the comparison arena. Their failure is not descriptive failure. Their failure is primitive identity failure.

Table 2: Representative identity catalogue

Candidate	What it can lawfully do	Failed criterion	Why it fails as primitive identity ground	Allowed status
Scalarity	Present Higgs-sector degrees of freedom in scalar-field language.	C2, C5	Scalar form alone does not supply gauge-to-deformation mediation, mass-response standing, or identity-role occupation.	Representative of C_H .
Vev-status	Coordinate electroweak symmetry-reduction and input normalization.	C4, C5	A vev coordinate depends on the admitted bridge context and cannot select the identity role it describes.	Boundary-normalized or witness-derived representative.

Candidate	What it can lawfully do	Failed criterion	Why it fails as primitive identity ground	Allowed status
Potential minimum	Present a model chart, symmetry-breaking profile, or diagnostic shape.	C3, C5	Potential charts are not identity grounds without quotient-stable descent through the carrier and admissibility boundary.	Diagnostic/model representative.
Pole mass	Record measured Higgs excitation response.	C4, C5	The pole mass is downstream response content; it measures an excitation associated with the carrier but does not create identity.	Empirical response witness.
Running/RG label	Transport scale-dependent parameters and diagnostics.	C3, C5	RG chart values depend on transport package and cannot supply primitive identity standing.	Transport representative.
Yukawa form	Express compatibility of fermion mass response with the Higgs-domain carrier.	C4, C5	Yukawa terms presuppose bridge admissibility and mass-response readiness.	Downstream compatibility route.
Mass matrix	Encode downstream deformation-response structure after bridge and Yukawa standing.	C4, C5	Mass matrices are not available as identity grounds before carrier standing.	Downstream response object.
Vacuum branch	Describe stability, metastability, or lower-branch route status.	C1, C5	Vacuum routes are endpoint statuses indexed to the carrier, not carrier identity itself.	Downstream route status.
Thermal transition	Describe finite-temperature route or electroweak transition status.	C1, C5	Thermal history does not supply the zero-temperature carrier identity and may require a distinct target.	Downstream thermodynamic route.
Metric comparison	Project carrier relations into a metric or hierarchy display.	C4, C5	Metricity is a projection surface; it preserves or displays carrier order but does not ground Higgs identity.	Downstream projection.

Candidate	What it can lawfully do	Failed criterion	Why it fails as primitive identity ground	Allowed status
Extension content	Add scalar, composite, portal, or richer-sector structure.	C1, C2, C5	Added structure either preserves C_H , changes scope, or fails a role; it is not same-scope identity by similarity alone.	Extension classifier input.

Lemma 4.1 (Representative failure certificate). *Each representative in Table 2 fails at least one identity-ground criterion unless its use factors through C_H or declares target update.*

Proof. Scalarity lacks carrier priority and does not by itself fix deformation mediation. Vev-status lacks selector neutrality if used to choose the carrier. Potential minima depend on a chart and must descend through a standing quotient. Pole mass is response witness, not carrier. RG labels are transport representatives. Yukawa forms and mass matrices presuppose bridge and mass-response admissibility. Vacuum branches and thermal transitions are route statuses with their own verdict maps. Metric comparison is a projection surface. Extension content changes or enriches the target unless it preserves C_H . In every case, primitive identity use either fails a criterion or is lawful only after factorization through the carrier. $\square$

Theorem 4.2 (Primitive Higgs identity collapse). *The familiar representative candidates do not survive as primitive same-scope Higgs identity grounds:*

$$\begin{aligned} \text{Surv}_{D_H}^{\text{Primitive}}(C_{\text{scalar}}) &= \text{Surv}_{D_H}^{\text{Primitive}}(C_v) = \text{Surv}_{D_H}^{\text{Primitive}}(C_V) \\ &= \text{Surv}_{D_H}^{\text{Primitive}}(C_{m_H}) = \text{Surv}_{D_H}^{\text{Primitive}}(C_{\text{RG}}) = 0. \end{aligned}$$

and likewise for Yukawa-form, mass-matrix, vacuum-branch, thermal-transition, metric-comparison, and extension-content candidates when used as same-scope primitive identity grounds.

Proof. By the identity-ground criterion, a primitive identity ground must preserve D_H , preserve H_{role} , remain invariant under identity-preserving redescription, avoid hidden selector or route-success authority, and supply identity-role occupation. The representative failure certificate shows that every listed representative fails one of these conditions unless its standing-bearing use factors through C_H or exits to $D_{H,+}$. Factorization through C_H means the representative is not the primitive identity ground; target update means the claim is no longer same-scope. Therefore no listed representative has a same-scope primitive identity survivor. $\square$

Corollary 4.3 (Standard Higgs descriptions are preserved). *The collapse theorem does not remove scalar-field, vev, potential, pole-mass, RG, Yukawa, mass-matrix, vacuum, thermal, metric, or extension descriptions. It removes only their promotion into primitive identity grounds.*

Proof. The theorem assigns each description an allowed status. Descriptions remain valid when they factor through the carrier, serve as route data, record empirical response, or declare target update. The deleted claim is primitive identity authority without carrier factorization. $\square$

5 Higgs identity normal form

With representative identity grounds deleted, identity-bearing use requires a normal form.

Definition 5.1 (Identity-specific normal form). The Higgs identity normal form is

$$R_H^{\text{id}} = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H^{\text{id}}, \rho_H^{\text{id}}),$$

where $\sim_H^{\text{id}}$ is identity-preserving same-scope redescription equivalence and ρ_H^{id} is the ordered fail-closed identity-status classifier.

Definition 5.2 (Identity-preserving redescription equivalence). For two Higgs identity presentations P, Q , write

$$P \sim_H^{\text{id}} Q$$

iff they preserve the same target D_H , the same carrier occupation C_H , the same role bundle H_{role} , the same representative/route status assignments, and the same admissibility boundary, up to notation, chart, field normalization, scheme, or presentation changes that do not alter standing-bearing identity content.

Definition 5.3 (Ordered fail-closed identity classifier). For a Higgs identity candidate X , the identity classifier $\rho_H^{\text{id}}(X)$ is assigned by first applicable status in the priority order

$$\begin{aligned} &\text{TargetUpdate} \succ \text{StandingSmuggling} \succ \text{RoleIncomplete} \\ &\succ \text{RouteStatus} \succ \text{RepresentativeOnly} \succ \text{CarrierIdentical}. \end{aligned}$$

The statuses are:

1. **TargetUpdate**: the candidate changes carrier inventory, target scope, admissibility boundary, or role bundle.
2. **StandingSmuggling**: the candidate treats representative availability, measurement success, route availability, empirical fit, or later extension as identity standing before carrier closure.
3. **RoleIncomplete**: the candidate preserves the target label but fails at least one component of H_{role} .
4. **RouteStatus**: the candidate is a downstream scale, vacuum, thermal, mass-response, hierarchy, or extension route rather than identity ground.
5. **RepresentativeOnly**: the candidate supplies descriptive representative content but no identity-role occupation.
6. **CarrierIdentical**: the candidate preserves D_H , H_{role} , and C_H under $\sim_H^{\text{id}}$.

Remark 5.4 (Priority order). The priority order makes the classifier functional. A target-changing representative is a target update, not merely representative-only. A route that treats itself as identity standing before closure is standing-smuggling. A representative-containing carrier presentation is not representative-only; it is carrier-identical if the carrier and role bundle are preserved.

Theorem 5.5 (Identity classifier totality). *Every submitted Higgs identity candidate receives exactly one status under the ordered fail-closed classifier ρ_H^{id} .*

Proof. The priority order first removes target updates, then standing-smuggling uses, then role-incomplete candidates, then downstream routes, then representative-only descriptions. If none of those statuses applies, the candidate preserves the target, admissibility boundary, role bundle,

Table 3: Identity classes and fixed-domain loci

Status class	Fixed-domain locus	Reason no further identity class remains
RepresentativeOnly	Standing partition	Descriptive availability is separated from identity standing.
RouteStatus	Admissible continuation	Downstream route status presupposes identity-bearing carrier occupation.
StandingSmuggling	Boundary typing / standing partition	Identity authority is imported before admissibility closure.
RoleIncomplete	Role-bundle preservation	Same-scope target label remains, but a required identity role is absent.
TargetUpdate	Boundary typing / target inventory	The candidate changes the domain rather than preserving same scope.
CarrierIdentical	Lawful quotient / admissible continuation	The candidate preserves the carrier role and survives redescription.

representative/route assignments, and identity-role occupation. That remaining case is **CarrierIdentical**. The priority order makes overlapping presentations functional, and Table 3 identifies the fixed-domain locus of each status. Hence every submitted candidate is classified exactly once. $\square$

Theorem 5.6 (Identity normal form theorem). *Every admissible same-scope standing-bearing Higgs identity ground factors through R_H^{id} :*

$$\begin{aligned} \text{IdClaim}_{D_H}(X) \wedge \text{SameScope}_{D_H}(X) \wedge \text{IdGround}_{D_H}(X) &= 1 \\ \Rightarrow \rho_H^{\text{id}}(X) &= \text{CarrierIdentical} \wedge \text{NF}_H^{\text{id}}(X) \downarrow \wedge X \sim_H^{\text{id}} \text{NF}_H^{\text{id}}(X), \end{aligned}$$

where $\text{NF}_H^{\text{id}}(X)$ is defined exactly in the **CarrierIdentical** case.

Proof. Let X be a same-scope standing-bearing Higgs identity ground. By classifier totality, $\rho_H^{\text{id}}(X)$ has exactly one status. Since X is same-scope and standing-bearing, it cannot be **TargetUpdate**, **StandingSmuggling**, **RoleIncomplete**, **RouteStatus**, or **RepresentativeOnly**: those statuses either leave the same target, fail standing, remain downstream, or supply description without identity-role occupation. The remaining status is **CarrierIdentical**. In that case the normalizer $\text{NF}_H^{\text{id}}(X)$ is defined by the preserved tuple $(D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H^{\text{id}}, \rho_H^{\text{id}})$, and $X \sim_H^{\text{id}} \text{NF}_H^{\text{id}}(X)$. Thus the normal form theorem applies exactly to admitted identity grounds, while rejected candidates remain classified by ρ_H^{id} . $\square$

Corollary 5.7 (Identity route availability is not identity closure). *A scale, vacuum, thermal, extension, or mass-response route may be available relative to C_H without becoming a primitive Higgs identity ground.*

Proof. Such a route is classified as **RouteStatus** unless it is merely representative or target-updating. The identity normal form reserves primitive identity standing for carrier-identical occupation. Route availability therefore does not close identity standing. $\square$

6 Uniqueness of role occupation

The identity theorem requires a stronger result than representative collapse. It must also show that a purported second same-scope occupant of the Higgs role is not a second identity ground.

Definition 6.1 (Same-scope Higgs replacement). A same-scope Higgs replacement is a candidate X such that X is asserted to occupy the Higgs identity role inside D_H , while preserving D_H , the admissibility boundary, and the role bundle H_{role} .

Definition 6.2 (Role occupation equivalence). For two candidates X, Y , write

$$X \simeq_{\text{role}} Y$$

iff X and Y perform the same bridge/fixation carrier work on D_H : they preserve gauge inheritance, residual/vacuum support, deformation mediation, mass-response readiness, identity continuity, standing transfer, redescription stability, and the same admissibility boundary.

Lemma 6.3 (Role-bundle exhaustion for identity work). *Inside the inherited target D_H , the identity-relevant carrier work is exhausted by preservation of D_H , H_{role} , identity-preserving redescription, representative/route status assignments, and the admissibility boundary. An asserted same-scope identity occupant that preserves all of these fields performs no additional primitive identity work outside the Paper-2 normal form.*

Proof. Paper 1 supplies D_H , C_H , H_{role} , $\mathcal{R}_H$, R_H , and SF_H as the local carrier package. Paper 2 refines this package by adding $\sim_H^{\text{id}}$ and ρ_H^{id} . The classifier totality theorem assigns every submitted candidate to target update, standing-smuggling, role incompleteness, route status, representative-only status, or carrier-identical status. The fixed-domain loci in Table 3 leave no additional identity-bearing location. Therefore any same-scope candidate preserving the target, role bundle, redescription equivalence, representative/route assignments, and admissibility boundary performs exactly the identity-relevant carrier work recorded by R_H^{id} . $\square$

Lemma 6.4 (Same-scope replacement lemma). *If X is a same-scope Higgs replacement and X preserves H_{role} , then*

$$X \simeq_{\text{role}} C_H.$$

Proof. Paper 1 established C_H as the closed bridge/fixation carrier occupying the role forced by D_H . A same-scope replacement preserving H_{role} performs the same governance and mediation work: gauge inheritance, residual/vacuum support, deformation mediation, mass-response readiness, identity continuity, standing transfer, and redescription stability. By role-bundle exhaustion for identity work, no further primitive identity work remains outside these preserved fields plus target and boundary preservation. If X differed in identity-relevant carrier work, it would fail a component of H_{role} , alter redescription status, or change target. If it does not differ in this work, it is role-equivalent to C_H . Thus $X \simeq_{\text{role}} C_H$. $\square$

Lemma 6.5 (Role equivalence induces identity redescription). *If $X \simeq_{\text{role}} C_H$ while X preserves D_H , the representative/route status assignments, and the admissibility boundary, then*

$$X \sim_H^{\text{id}} C_H.$$

Proof. By definition, $X \simeq_{\text{role}} C_H$ preserves the bridge/fixation carrier work: gauge inheritance, residual/vacuum support, deformation mediation, mass-response readiness, identity continuity, standing transfer, redescription stability, and the admissibility boundary. Together with preservation of D_H and the representative/route status assignments, these are exactly the fields required by identity-preserving redescription equivalence. Hence $X \sim_H^{\text{id}} C_H$. $\square$

Theorem 6.6 (No second same-scope Higgs identity ground). *There is no same-scope primitive Higgs identity ground G such that*

$$\text{IdGround}_{D_H}(G) = 1, \quad G \not\simeq_{\text{role}} C_H,$$

while G preserves D_H and H_{role} .

Proof. Assume such a G exists. Since G preserves D_H and H_{role} , it is a same-scope Higgs replacement. By the same-scope replacement lemma, $G \simeq_{\text{role}} C_H$. This contradicts $G \not\simeq_{\text{role}} C_H$. Therefore no second same-scope primitive identity ground exists. $\square$

Corollary 6.7 (Same role, different presentation). *A candidate that preserves all identity roles while using a different scalar representation, field normalization, gauge chart, potential presentation, compositeness language, or extension vocabulary is not a second Higgs identity. It is the same carrier role under redescription, unless it changes target.*

7 No fifth identity ground

The preceding sections collapse representatives and same-scope replacements. A final counterexample form remains: an unlisted identity ground.

The word “fifth” does not count the six fail-closed classifier statuses. It refers to the fixed-domain no-fifth-locus discipline inherited from Paper 1: after boundary typing, standing partition, lawful quotient, admissible continuation, and target update have been accounted for, no additional same-scope governance-bearing identity locus may be introduced by naming it.

Definition 7.1 (Fifth identity function). A fifth identity function is a map $F_5(X)$ asserted to ground Higgs identity while satisfying all of the following:

1. $F_5(X) \not\equiv \text{Occ}(C_H)$;
2. $F_5(X)$ is not representative-only;
3. $F_5(X)$ is not a downstream route status;
4. $F_5(X)$ is not a target update;
5. $F_5(X)$ does not import standing by smuggling or repair;
6. $F_5(X)$ preserves D_H , H_{role} , and the admissibility boundary.

Criterion 7.2 (Standing test for a fifth identity ground). A fifth identity function has standing only if it supplies same-scope identity authority while preserving D_H , preserving H_{role} , avoiding representative promotion, avoiding route promotion, avoiding standing-smuggling, avoiding target update, and not reducing to C_H under role occupation equivalence.

Theorem 7.3 (No fifth identity ground). *No fifth same-scope Higgs identity ground satisfies the standing test. Equivalently,*

$$\neg \exists F_5 \text{ Stand}_{D_H}^{\text{prim}}(F_5) = 1.$$

Proof. Let F_5 be asserted. If it supplies descriptive content, it is representative-only. If it supplies scale, vacuum, thermal, extension, mass-response, or hierarchy content, it is route status. If it treats such content as identity before carrier closure, it is standing-smuggling. If it changes carrier inventory, role bundle, or admissibility boundary, it is target update. If it preserves the target and full role bundle while grounding identity, then by the same-scope replacement lemma it is role-equivalent to C_H , contradicting the condition $F_5 \not\equiv \text{Occ}(C_H)$. These alternatives exhaust the fixed-domain identity loci. Therefore no fifth same-scope identity ground remains. $\square$

Corollary 7.4 (Fifth-ground closure entails structural identity). *Once representative, route, smuggling, incomplete-role, and target-update alternatives are closed, carrier occupation is the unique same-scope identity ground.*

8 Endpoint theorem

Theorem 8.1 (Higgs Structural Identity Theorem). *In the fixed Higgs-domain target D_H with inherited carrier package $\mathcal{P}_H^{(1)}$, Higgs identity is unique admissible bridge/fixation carrier occupation:*

$$\text{Id}_H = \text{Occ}_{D_H}(C_H).$$

Equivalently, for any candidate X ,

$$\text{IdGround}_{D_H}(X) = 1 \iff X \sim_H^{\text{id}} C_H,$$

provided the claim is same-scope and preserves H_{role} .

Proof. By Paper 1, C_H is the standing-bearing bridge/fixation carrier forced by the Higgs-domain target and closed by the imported bridge-witness package. Therefore $\text{Occ}_{D_H}(C_H)$ supplies a same-scope identity ground. Conversely, let X be any same-scope identity ground preserving H_{role} . If X is representative-only, route-status, standing-smuggling, role-incomplete, or target-updating, it is not a same-scope primitive identity ground. If it is none of those, the identity normal form theorem assigns it CarrierIdentical status, and the same-scope replacement lemma gives $X \simeq_{\text{role}} C_H$. Since X is same-scope and carrier-identical, it preserves D_H , the representative/route assignments, and the admissibility boundary. By the role-equivalence-to-redescription lemma, $X \sim_H^{\text{id}} C_H$. Hence Higgs identity is unique admissible occupation of C_H . $\square$

Corollary 8.2 (Primitive identity survivor). *The only Paper-2 same-scope primitive identity survivor is C_H :*

$$\text{Surv}_{D_H}^{\text{Primitive}}(C_H) = 1,$$

while all primitive representative identity claims have survivor value zero.

Corollary 8.3 (Identity before scale, vacuum, and extension). *Higgs scale, vacuum, thermal, and extension claims are evaluated downstream of $\text{Id}_H = \text{Occ}_{D_H}(C_H)$. They cannot determine Higgs identity in the same scope.*

9 Relation to Standard Model Higgs language

The theorem preserves ordinary Standard Model Higgs language by typing it correctly. The Standard Model scalar doublet, Higgs potential, vev convention, pole-mass measurement, Yukawa couplings,

and electroweak mass relations remain legitimate representational and empirical structures. The theorem says that their identity-bearing use is mediated by the carrier role.

Proposition 9.1 (Scalar doublet presentation). *A scalar doublet presentation is a lawful representative of the Higgs carrier when embedded in the Standard Model structural target. It is not a primitive identity ground independent of the bridge/fixation role.*

Proof. The scalar doublet presentation can encode gauge representation, symmetry reduction, and interaction structure. Those are precisely descriptions of the Higgs-domain carrier in a licensed Standard Model presentation. But scalarity by itself does not supply deformation mediation, mass-response readiness, identity continuity, or standing transfer. Therefore it is representative of C_H , not independent identity ground. $\square$

Proposition 9.2 (Measured Higgs excitation). *The Higgs pole mass and excitation records are empirical response witnesses of the carrier. They do not select the carrier.*

Proof. A pole mass is measured only after the identity target of the excitation is fixed. It bears empirical response content, but it cannot by itself determine what the measured object is as a same-scope Higgs identity ground. Its identity-bearing use presupposes C_H . $\square$

Proposition 9.3 (Vacuum and thermal descriptions). *Vacuum and thermal Higgs claims are downstream route statuses. They may affect endpoint classification, stability classification, or transition classification, but not primitive Higgs identity.*

Proof. Vacuum and thermal claims are evaluated by route-specific targets: static/lifetime/thermal verdicts, matching and RG transport, and possible richer-scope updates. Those routes presuppose a Higgs carrier whose potential, branch, or thermal history is being evaluated. They do not create the carrier identity. $\square$

10 Downstream interfaces

The identity endpoint supplies formal inputs to the subsequent papers in the Higgs structural endpoint arc.

Status 10.1 (Scale interface). A Higgs scale claim in the next scale paper is not a primitive identity claim. It must be evaluated as a carrier-indexed scale representative or carrier-order projection:

$$\text{ScaleClaim}_H(q) \Rightarrow q \text{ is indexed to } C_H \text{ or target-updating.}$$

Status 10.2 (Vacuum interface). A Higgs vacuum claim is not a primitive identity claim. It is a route status indexed to C_H , governed by static, lifetime, thermal, or richer-scope verdict maps.

Status 10.3 (Extension interface). An extra scalar, composite construction, second doublet, or richer-sector construction is not a same-scope Higgs identity merely by scalar similarity, vev participation, or coupling. It must preserve C_H , duplicate a representative, split the role under a richer-scope declaration, or fail.

11 Claim and nonclaim register

Table 4: Paper-2 claim and nonclaim register

ID	Statement	Status
HID-T1	Paper 2 works downstream of the Paper-1 carrier package.	Dependency lock
HID-T2	A Higgs identity claim is stronger than a Higgs description.	Definition / theorem spine
HID-T3	Descriptive adequacy does not imply identity-grounding standing.	Lemma
HID-T4	Scalarity, vev-status, potential-minimum status, pole-mass status, RG status, Yukawa-form status, mass-matrix status, vacuum-branch status, thermal-transition status, and metric-comparison status fail as primitive identity grounds.	Collapse theorem
HID-T5	Identity-role occupation is neutral at admission and does not select C_H by definition.	Anti-circularity lemma
HID-T6	Every submitted Higgs identity candidate receives exactly one fail-closed identity status.	Classifier totality theorem
HID-T7	Every same-scope standing-bearing Higgs identity ground factors through R_H^{id} .	Normal-form theorem
HID-T8	Same-scope replacements preserving H_{role} reduce to C_H , and role equivalence induces identity redescription.	Replacement theorem
HID-T9	No fifth same-scope identity ground remains.	Exhaustion theorem
HID-T10	Higgs identity is unique admissible occupation of C_H .	Main endpoint theorem
HID-N1	The paper does not deny the scalar doublet representation.	Nonclaim
HID-N2	The paper does not compute Higgs numerical quantities.	Nonclaim
HID-N3	The paper does not rule out richer-scope Higgs-sector extensions.	Nonclaim
HID-N4	The paper does not classify all BSM models.	Nonclaim

12 Denial obligations and same-target counterexamples

A same-target denial of the paper must do at least one of the following:

1. break the inherited carrier package $\mathcal{P}_H^{(1)}$;
2. show that a Higgs identity claim does not require preservation of D_H , H_{role} , and the admissibility boundary;
3. show that identity-role occupation smuggles C_H into the admission criterion;

4. supply a representative that independently satisfies the identity-ground criterion without factoring through C_H ;
5. show that the identity classifier leaves a same-scope governance-bearing class unclassified;
6. show that a rejected same-scope candidate must receive normal-form factorization rather than a closed status;
7. supply a same-scope replacement preserving H_{role} but not role-equivalent to C_H ;
8. break the implication from role equivalence plus target and boundary preservation to identity redescription equivalence;
9. supply a fifth identity function satisfying the fifth-ground standing test;
10. show that target update is actually same-scope while changing carrier inventory or the admissibility boundary.

Theorem 12.1 (Same-target counterexample burden). *A faithful same-target counterexample must preserve D_H , C_H 's inherited role closure, H_{role} , identity-preserving redescription, the admissibility boundary, and the representative/route distinction, while producing a primitive identity ground not role-equivalent to C_H .*

Proof. If the counterexample changes target, role bundle, carrier closure, redescription equivalence, or admissibility boundary, it is a target update or scope change. If it uses a representative or route as identity authority, it is representative promotion or standing-smuggling. If it preserves all fixed items and supplies identity standing, then by the same-scope replacement lemma it is role-equivalent to C_H . Therefore a same-target counterexample must preserve all fixed items while violating the replacement lemma or no-fifth-ground theorem. $\square$

13 Anti-circularity and target non-smuggling

The identity endpoint is not obtained by defining C_H to be the Higgs. The manuscript uses three distinct steps: carrier admission, identity-claim typing, and endpoint exhaustion. Only the third step states the identity theorem.

Proposition 13.1 (Target neutrality). *The inherited target D_H does not contain the conclusion $\text{Id}_H = \text{Occ}_{D_H}(C_H)$. It contains the arena in which identity claims are evaluable: order-to-deformation transition, mass-response readiness, role-bundle preservation, representative catalogue, and same-scope update discipline.*

Proof. A target that contains an arena, candidate representatives, and update conditions does not by itself select an identity ground. D_H admits scalar, vev, potential, mass, RG, vacuum, thermal, and extension language as candidate identity or representative material. The identity conclusion is reached only after the representative-collapse theorem, identity normal-form theorem, same-scope replacement lemma, and no-fifth-ground theorem are applied. Hence the conclusion is not present in the target definition. $\square$

Proposition 13.2 (Carrier admission is not identity closure). *The Paper-1 result $C_H = H_{\text{br/fix}}$ admits a carrier. It does not, by itself, prove the Paper-2 identity endpoint.*

Proof. Carrier admission states that a bridge/fixation carrier is forced by the Higgs-domain transition target and that the closed occupant is C_H . Identity closure requires an additional statement: every same-scope Higgs identity claim must either reduce to that carrier occupation, remain representative, become route-status, fail, smuggle standing, or update target. Those alternatives are not exhausted merely by naming C_H . They are exhausted by the classifier and no-fifth-ground theorem in this paper. $\square$

Table 5: Anti-circularity checkpoints

Potential circularity	Guardrail in manuscript	Resulting status
D_H defines the Higgs as C_H .	D_H is an evaluability target, not an identity selector.	No target smuggling.
C_H is declared identical to the Higgs by notation.	C_H is imported as carrier; identity follows only after representative and fifth-ground exhaustion.	No naming fiat.
Representatives are excluded by definition.	Representatives are admitted and assigned lawful statuses.	No representative erasure.
Extensions are ruled out globally.	Extensions are classified as carrier-preserving, representative, role-splitting, target-updating, or failing.	No anti-BSM overclaim.
Routes are dismissed.	Routes are preserved as downstream statuses.	No route denial.

14 Endpoint theorem set

The preceding results can be collected as a compact endpoint package.

Theorem 14.1 (Identity-description separation theorem). *A same-scope Higgs description is not a Higgs identity ground merely by accurately describing a Higgs-associated object.*

Proof. This is the identity-description separation lemma applied to the Higgs-domain representative catalogue. Accuracy of description is compatible with representative-only status, route-status, or empirical witness status. Identity-grounding standing requires carrier occupation and preservation of H_{role} . $\square$

Theorem 14.2 (Representative identity deletion theorem). *Every primitive identity claim grounded only in scalarity, vev-status, potential-minimum status, pole-mass status, RG-running status, Yukawa-form status, mass-matrix status, vacuum-branch status, thermal-transition status, metric-comparison status, or extension similarity is deleted in D_H .*

Proof. Each such claim is one row of the representative catalogue. The failure certificate for the row identifies the missing identity-ground component. If the component is supplied by C_H , the representative is no longer primitive. If it is supplied by new carrier content, the claim updates the target. Therefore primitive same-scope identity standing is deleted. $\square$

Theorem 14.3 (Identity factorization theorem). *Every same-scope standing-bearing Higgs identity ground factors through R_H^{id} .*

Proof. This is the identity normal form theorem restated as an endpoint package: admitted same-scope identity-bearing use must preserve the target, role bundle, admissibility boundary, representative classification, and identity redescription equivalence. Those are exactly the fields of R_H^{id} . Submitted candidates that do not have identity-ground standing remain classified by ρ_H^{id} , but they do not receive normal-form factorization. $\square$

Theorem 14.4 (Replacement uniqueness theorem). *Every same-scope Higgs replacement preserving H_{role} is role-equivalent to C_H .*

Proof. This is the same-scope replacement lemma. A replacement that preserves all role work performs the same carrier work. A replacement that does not preserve all role work fails or changes target. $\square$

Theorem 14.5 (Identity endpoint registry theorem). *The Paper-2 endpoint registry is*

$$\text{Reg}_{\text{id}}(D_H) = (\text{carrier identity closed, representatives non-primitive,} \\ \text{routes downstream, no fifth ground}).$$

It does not include numerical Higgs values, vacuum selection, hierarchy magnitude, or BSM exclusion.

Proof. The registry records exactly the theorem outputs proved above. Numerical values, vacuum verdicts, scale magnitudes, and BSM classifications require additional target-specific certificates and are not part of the Paper-2 identity endpoint. $\square$

15 Weakening resistance

A same-target weakening preserves the Higgs identity target while weakening one of the conditions that makes the identity endpoint possible. The following table records the cost of each weakening.

Table 6: Cost of weakening the identity structure

Weakened condition	Immediate loss	Effect on identity theorem
Target fixation D_H	The object of identity evaluation drifts among scalar, vev, mass, route, and extension claims.	No fixed identity claim remains.
Carrier occupation C_H	Representatives can be used as identity grounds without role closure.	Standing collapses into description.
Role bundle H_{role}	Gauge inheritance, deformation mediation, mass response, or standing transfer becomes optional.	Same-scope Higgs identity loses its target.
Identity redescription $\sim_H^{\text{id}}$	Presentation changes can create or destroy identity.	Identity becomes chart-dependent.

Weakened condition	Immediate loss	Effect on identity theorem
Representative/route distinction	Routes and measurements can be treated as primitive.	Standing-smuggling becomes possible.
Target-update discipline	New carrier content can be introduced while still called same-scope.	Scope repair becomes uncontrolled.
No-fifth-ground closure	Unspecified classifiers can be asserted after exhaustion.	The endpoint is reopened by naming, not proof.

Theorem 15.1 (No strict same-target weakening). *No strict weakening preserves the Paper-2 identity target while admitting a primitive representative identity ground or a fifth identity ground.*

Proof. If target fixation is weakened, the object of evaluation changes. If carrier occupation is weakened, representative descriptions become unauthorized identity grounds. If H_{role} is weakened, the identity target no longer preserves the conditions that make a Higgs-domain carrier evaluable. If redescription equivalence is weakened, identity is presentation-dependent. If route and representative distinctions are weakened, standing-smuggling is allowed. If target-update discipline is weakened, scope repair is allowed. In each case the target phenomenon is lost or a governance-equivalent condition is reintroduced. Hence no strict same-target weakening both preserves the target and admits the forbidden primitive. $\square$

16 Right-mode objection targets

A serious objection must attack a load-bearing theorem link rather than a preferred presentation. The right-mode targets are:

1. the inherited Paper-1 carrier package;
2. the definition of Higgs identity claim;
3. the identity-ground criterion;
4. the identity-role-occupation neutrality lemma;
5. the identity-description separation lemma;
6. the representative failure certificates;
7. the primitive identity collapse theorem;
8. the identity classifier totality theorem;
9. the identity normal form theorem;
10. the same-scope replacement lemma;
11. the role-bundle exhaustion and role-equivalence-to-redescription lemmas;
12. the no-second-identity-ground theorem;
13. the no-fifth-identity-ground theorem;
14. the endpoint theorem;
15. the same-target counterexample burden.

A statement that the Higgs is commonly represented by a scalar field does not attack any of these links. The paper grants that representation. A statement that an extension may contain additional scalar physics also does not attack the theorem; it enters the extension interface and must be classified there.

17 Conclusion

Paper 1 established the Higgs bridge/fixation carrier. Paper 2 closes the identity consequence. The Higgs-domain primitive is not scalarity, vev-status, a potential minimum, pole mass, running label, Yukawa form, mass matrix, vacuum branch, thermal history, metric comparison, or extension content. Those objects are legitimate only as representatives, empirical witnesses, route statuses, projections, enrichments, or target updates. The same-scope identity ground is unique admissible occupation of the bridge/fixation carrier:

$$\text{Id}_H = \text{Occ}_{D_H}(C_H).$$

The result is structural and status-theoretic. It preserves standard Higgs descriptions while preventing them from becoming primitive identity authorities. It also fixes the downstream order of the arc: scale, vacuum, and extension claims are not ways of discovering what the Higgs is; they are claims about attributes, routes, projections, or enrichments of the carrier whose identity has already been fixed.

A Identity representative catalogue

Table 7: Identity representative audit

Representative	Identity temptation	Failed criterion	Failure mode	Correct status
Scalar field	The Higgs is the scalar field.	C2, C5	Scalality is too weak; it does not supply full role bundle or identity-role occupation.	Presentation of C_H .
Higgs doublet	The Higgs is the doublet representation.	C2, C5	Representation is necessary in SM presentation but not self-grounding.	Structural representative.
Vev	The Higgs is the vev.	C4, C5	Vev presupposes carrier and normalization.	Boundary/witness representative.
Potential minimum	The Higgs is the minimum of V .	C3, C5	Potential chart does not itself fix identity.	Diagnostic/model representative.
Pole mass	The Higgs is the 125 GeV excitation.	C4, C5	Measurement identifies response after carrier identity.	Empirical witness.

Representative	Identity temptation	Failed criterion	Failure mode	Correct status
Running quartic	The Higgs is the RG-running scalar parameter.	C3, C5	Running is transport, not identity.	Transport diagnostic.
Yukawa terms	The Higgs is what gives fermion masses.	C4, C5	Yukawa compatibility presupposes bridge readiness.	Downstream compatibility.
Mass matrix	The Higgs is mass-generating matrix structure.	C4, C5	Matrix content is downstream deformation response.	Mass-response output.
Vacuum branch	The Higgs is its vacuum fate.	C1, C5	Vacuum fate is endpoint route, not identity ground.	Route status.
Thermal history	The Higgs is electroweak transition behavior.	C1, C5	Thermal target differs from identity target.	Thermodynamic route.
Metric hierarchy	The Higgs is a low metric scale.	C4, C5	Metric projection displays carrier relation.	Downstream projection.
Extra scalar	The Higgs is any scalar sharing the role.	C1, C2, C5	Similarity does not prove same carrier occupation.	Extension classifier input.

B Dependency map from Paper 1 to Paper 2

Table 8: Dependency map

Imported item	Paper-1 role	Paper-2 use
Kernel roles	Establish target, standing, reference, irreversibility.	Identity claims inherit same governance burden.
AMetric boundary	Blocks pre-gate selector, metric, ranking, or representative authority.	Blocks representative identity selectors.
Fixed-domain exhaustion	Closes no-fifth-locus governance variation and fixes classifier loci.	Supports classifier totality and no-fifth-identity-ground theorem.
Bridge-witness closure	Identifies $H_{\text{br/fix}}$ as closed occupant.	Supplies C_H for identity endpoint.
Representative collapse	Deletes primitive scalar, vev, potential, mass, RG, and vacuum use.	Extends collapse from standing to identity.
Normal form R_H	Factors admitted same-scope Higgs standing claims.	Specializes to R_H^{id} for admitted identity grounds.

Imported item	Paper-1 role	Paper-2 use
Standing filter $\mathbf{SF}_H$	Separates primitive, representative, route, failure, and update claims.	Specializes to identity classifier ρ_H^{id} and its normalizer boundary.

C Formal status machine

The identity classifier may be recorded as a status machine:

$$\begin{aligned}
\rho_H^{\text{id}}(X) = \mathbf{TargetUpdate} &\iff X \text{ changes } D_H \text{ or the boundary,} \\
\rho_H^{\text{id}}(X) = \mathbf{StandingSmuggling} &\iff X \text{ promotes representative or route availability,} \\
\rho_H^{\text{id}}(X) = \mathbf{RoleIncomplete} &\iff X \text{ fails at least one } H_{\text{role}} \text{ component,} \\
\rho_H^{\text{id}}(X) = \mathbf{RouteStatus} &\iff X \text{ supplies downstream route content,} \\
\rho_H^{\text{id}}(X) = \mathbf{RepresentativeOnly} &\iff X \text{ supplies descriptive content only,} \\
\rho_H^{\text{id}}(X) = \mathbf{CarrierIdentical} &\iff X \sim_H^{\text{id}} C_H.
\end{aligned}$$

Theorem C.1 (Classifier completeness). *Every submitted Higgs identity candidate receives exactly one status by the ordered fail-closed classifier.*

Proof. The priority order resolves overlap. A target-changing representative is **TargetUpdate**; a route that asserts identity authority is **StandingSmuggling**; a candidate missing a role is **RoleIncomplete**; a downstream claim is **RouteStatus**; a pure description is **RepresentativeOnly**; and the remaining case is carrier preservation. Since the status locations correspond to the fixed-domain loci of target boundary, standing partition, role-bundle preservation, admissible continuation, representative availability, and lawful quotient, no governance-bearing case remains outside the classifier. $\square$

Only candidates assigned **CarrierIdentical** are normalized by $\mathbf{NF}_H^{\text{id}}$. The other statuses are closed reports, not identity-normal-form survivors.

D Source-lock and support manifest

Table 9: Support source roles

Source family	Imported role	Paper-2 use
Paper 1 carrier manuscript	$D_H, C_H, H_{\text{role}}, \mathcal{R}_H, R_H, \mathbf{SF}_H, E_H^{(1)}$.	Direct dependency package and role-bundle exhaustion input.
Kernel paper	Four-role kernel and non-degenerate construction.	Already imported through Paper 1.
Structure of admissibility	AMetricity, bivalence, standing identity, unique interior, conservation.	Prevents identity-ground grading or representative selection.

Source family	Imported role	Paper-2 use
Fixed-domain exhaustion	No fifth governance-bearing locus and fixed-domain classifier locations.	Supports classifier totality and no fifth identity ground.
Closure by exhaustion	Same-scope operator closure and bookkeeping/inadmissible dichotomy.	Supports no emergent identity operator or unclassified survivor.
AMetric boundary	No selector, parameter, metric, canonical representative at boundary.	Blocks scalar/vev/potential identity selectors.
Impossibility suite	No independent generators, no transfers, no repairs.	Blocks route success and representative success as identity authority.
MGN3 Higgs bridge support	Bridge witness package, quotient-reduced carrier, role-rank.	Supplies bridge/fixation closure behind C_H .
Standard Model structural support	SM structural tuple and Higgs bridge/fixation role.	Locates carrier in electroweak structural arena.
Vacuum endpoint support	Branch and metastability route status.	Confirms vacuum claims are downstream routes.
Scale/hierarchy support	No primitive scale and metric projection discipline.	Confirms scale/metric claims are downstream projections.

E Objection register

Table 10: Objection register

Objection	Standing	Response
The Higgs is a scalar field.	Partial	Granted as representative description; denied as primitive identity ground.
The vev selects the Higgs.	None as identity selector	Vev-status presupposes bridge/fixation context and cannot select the carrier.
The measured pole mass identifies the Higgs.	Partial	The pole mass is an empirical response witness, not the identity ground.
The Standard Model already defines the Higgs.	Compatible	The paper classifies the standing role of that definition inside the structural target.
An extra scalar may be another Higgs.	Serious if carrier-preserving	Must preserve H_{role} and either reduce to C_H , enrich downstream structure, or declare target update.
Composite Higgs models replace the identity theorem.	Target-dependent	A composite route either preserves the same carrier role under different realization or changes scope.
Vacuum fate determines Higgs identity.	None	Vacuum fate is downstream route status indexed to the carrier.

Objection	Standing	Response
Metric hierarchy determines Higgs identity.	None	Metric hierarchy is downstream projection, not identity ground.
A fifth identity ground exists.	Serious if specified	Must satisfy the fifth-ground standing test while avoiding reduction to C_H .

F Notation index

Table 11: Notation index

Symbol	Meaning
D_H	Fixed Higgs-domain target inherited from Paper 1.
C_H	Higgs bridge/fixation carrier $H_{\text{br}/\text{fix}}$.
H_{role}	Role bundle required for Higgs-domain standing and identity.
$\mathcal{R}_H$	Catalogue of admitted Higgs representatives.
R_H	Local Higgs normal form from Paper 1.
R_H^{id}	Identity-specific normal form.
SF_H	Higgs standing filter from Paper 1.
ρ_H^{id}	Identity-status classifier.
$\sim_H^{\text{id}}$	Identity-preserving same-scope redescription equivalence.
$\simeq_{\text{role}}$	Role occupation equivalence.
Id_H	Higgs structural identity endpoint.
$\text{Occ}_{D_H}(C_H)$	Unique admissible occupation of the carrier role inside D_H .

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